

On the factors describing flowstate; A cognitive perspective

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1 Introduction

In an increasingly digital society, many teenagers and young adults find issues struggling to focus academically as distractions become ever more prevalent, many new terms such as "flow state" and the state of being "locked in" having increasingly used in social media platforms describe "flow", or, the mental state of total deep engrossment/absorption in a cognitive work and an ensuing loss of self-consciousness.(Mazid et al., 2024) This state is linked to an optimal improvement/learning schedule(Mazid et al., 2024). And hence, in the contemporary world, this is a particularly an important field of research as our target population of teenagers, given too many choices in life, would find it easier to engage in activities such as watching television, "doomscrolling", and gaming due to the lack of pressure for teenagers to doing "challenging activities that can teach and improve new skills" (Schwartz, 2000) (Mazid et al., 2024). In this stage in life according to piaget's theory of development teenagers are transitioning into higher-order thinking and requires a lot of heavy learning to occur, hence, it is important for them to engage in skill learning as it is a critical developmental stage in human life. 'Being in the state of flow meant having an optimal experience which has been referred to as "the bottom line of existence

(because) without it there would be little purpose in living.”.’ (Mazid et al., 2024)

Modern research has identified 9 characteristics of flow the flow experience. The challenge skill balance is often considered the backbone of flow research. This characteristic suggests that flow is reached when the perceived challenge of the task matches the skill level of the participant. Illustrated in figure 1. This parallels to Vygotsky’s ”Zone of Proximal Development” which is being in a skill level where it requires help from a more knowledgeable other rather than being impossible nor able to be done without any external assistance.

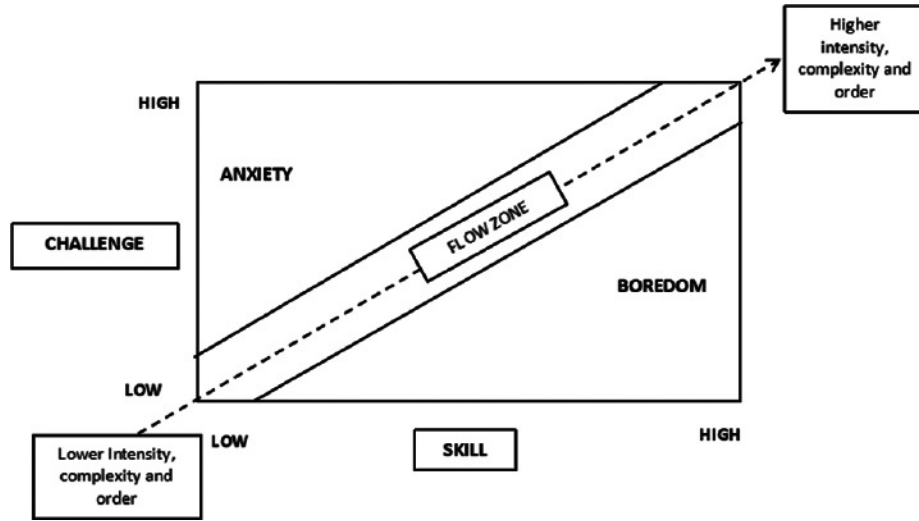


Figure 1: The challenge-skill balance model of flow state (Mazid et al., 2024)

However, many other factors were later discovered in relating to flow, such as perceived control. However, this does not necessarily mean the advancement in resources. For instance, giving one a calculator in a math test, may increase their ”perceived control”, but that is because it does really increase their control in their situation, similar to having access to more money/funding to accomplish a task. And it is imperative that we operationalize perceived control carefully.

Hence, this research will focus on the effect of perceived control on the magnitude of flow measured as a classification based on absorption, timelessness and effort.

2 Methodology

2.1 Sampling Population

Our population will be chosen based on opportunity sampling. We will choose 10th grade students based on their grades in humanities class since 10 grade

students are relatively blithe compared to us IB die-hards, which satisfies the criterion for low pressure for work, meaning they just so happen to have more reason to need to improve their skills in this stage of development. According to Csikszentmihalyi's model, the skill-challenge balance plays an important role in achieving flow state, therefore it is important that we establish a population of similar skill level, and the challenge will be the same as we will maintain same procedure for each trial. It is known last year that we were "voluntold" to be subject to the senior's dreadful experiments to get them their diploma. A similar setup should be possible (had the ib maintained allowing psychological experimenting and) if our school were open to coordination. Hence, in order for the skill level to stay constant based on Csikszentmihalyi's model we would choose a sample population based on stratified sampling on the same humanities class. It is possible to obtain access to their grades having expressed our dire situation. If rejected like college applications, we may resort to systemic sampling by engaging in the ethically illicit activity known as being familiar with students in their grade to know who are the high-achieving students and who are not and then stratifically sample based on the best of our ability. [yeah definitely need work on wording that]

2.2 Ethical Considerations

It will be disclosed to each participant that at any point in the study they have the right to withdraw, informing that their data is to be deleted in such cases.

As this is noted, we will provide a consent form that will include all potential negative effects of such experimenting. Which includes:

- Pressure to complete the task (may exist in high-achieving students)
- Slight embarrassment (from watching yourself, but this will not be disclosed on the form such that they will not be distracted from the task at hand[but this whole part should be worded better])

It is known that all Psychological experimenting should attempt to result in maximising benefit and minimizing harm, referred to as "beneficence". This suggests that our independent variable should be unipolar, controlling the magnitude of perceive control rather than purely depriving them of control. This would theoretically only result in effect would be varying magnitudes of improvement rather than any deprovements.

Minor levels of deception will be used, since we will record their screen during work as well as provide a facecam to refer to afterwards. This will be reduced to "The program will be collecting data during work" without informing them of face/screen recording such that they may focus on the task at hand.

2.3 Changing the independent variable

The independent variables of "perceived control" has nuance, in one hand, it is their ability to choose what prompts to work on or what they are most

comfortable with, but what if they were unfamiliar with any of them. They would end up in (there was an experiment on mind field youtube that was like if someone choose it for you you would feel like you did better, but if you choose yourself you would feel worse, I'm certain there is a technical name for that)

2.4 Designing the Data Collection

Many methods of data collection for flow has been used such as the four-channel model ratio of challenge and skill, and a classification based on ratings of absorption, timelessness and effort (Ainley et al., 2008). However, we will focus on using a classification based on absorption, timelessness and effort. Since this better describes the effects of flow rather than the theoretical preconditions that affect flow such as skill-challenge balance. For instance, being in flow, one would certainly feel absorbed in the task, as per the definition of flow. However, it does not mean that this is inherently caused by a equal balance of challenge and skill. After all, we are interested in optimizing the beneficial effects of flow, hence, we must resort to the definition of the effects of flow rather than the theoretical preconditions of flow.

Operationalized variables: The level of flow should be experimentally determined by the following factors Potential speculated factors affecting it

The tool itself:

1. questionnaire/survey
2. experiment
3. observation
4. interview

you can only do 5 items anyways

the explanation of the decisions explain, not describe it has to be effective, valid, and reliable your tool needs to match the population. it has to be relevant to the population

Survey

potential challenges

I think I will be doing experiment, doing an assignment, throughout different times of the day. and recording factors such as food consumption, what you were doing earlier (productive/relaxing work), etc. And then seeing what factors most visibly affected the flowstate or if they even were able to achieve flowstate we use a standardized test I forgot what it was called but I am certain there are many ways of doing it. if they didn't achieve flowstate, explain why they didn't

3 Discussion

Discussion of the potential findings of the investigation and the implications for policyt/practice how it can potentially be used for making policies of the school for instance. *start with a paragraph bias:

1. how might've bias influenced
2. your personal history (are you involved?)

each criteria/section should have 3 paragraphs

potential findings are described in detail, and implications for policy/practice
relevant examples of how researcher bias is influenced (one is enough, I can
talk about confirmation bias)

one relevant additional research method with reference to increasing the
understanding of the area of investigation

References

- Ainley, M., Enger, L., & Kennedy, G. (2008). The elusive experience of 'flow': Qualitative and quantitative indicators [Application of qualitative and quantitative methods to enrich understanding of emotional and motivational aspects of learning]. *International Journal of Educational Research*, 47(2), 109–121. <https://doi.org/https://doi.org/10.1016/j.ijer.2007.11.011>
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